1 **(a)** Elastic collision: The **total** kinetic energy (KE of the system) of the colliding bodies is conserved before and after the collision.

Or The relative speed of approach of the two bodies is equal to their relative speed of separation.

(b) By conservation of linear momentum,

$$m_A u_A + m_B u_B = m_A v_A + m_B v_B$$

$$(2.4) + 0 = v_A + 4v_B \quad [\text{C1: sub } u_p \text{ and } m_B = 4m_A]$$

By conservation of kinetic energy

$$\frac{1}{2}m_A u_A{}^2 + \frac{1}{2}m_B u_B{}^2 = \frac{1}{2}m_A v_A{}^2 + \frac{1}{2}m_B v_B{}^2$$

$$(2.4)^2 + 0 = v_A{}^2 + 4v_B{}^2 \quad [\text{C1: sub } u_A \text{ , } m_B = 4m_A]$$

By relative speeds

$$u_A - u_B = v_B - v_A$$

$$(2.4) - 0 = v_B - v_A \quad [\text{C1: sub } u_p]$$

Note: only two C1 marks available
Solving simultaneously with any of the two above equations,

$$v_A = -1.44 \text{ m s}^{-1} \quad [\text{A1}]$$

(c) By conservation of linear momentum,
The total momentum of a system of objects remains constant provided no resultant external force acts on the system.

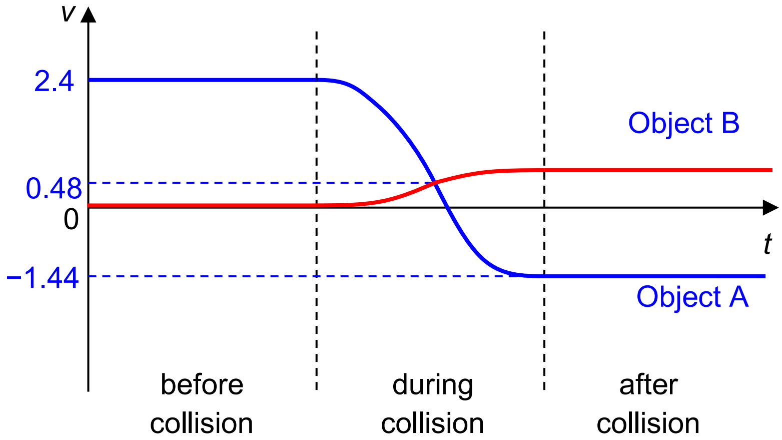
Since no external resultant force is acting on the system,

$$m_A u_A + m_B u_B = m_A v_o + m_B v_o$$

$$(2.4) + 0 = 5v_o \quad [\text{B1: sub } u_p \text{ and } m_{He} = 4m_p]$$

$$v_o = 0.48 \text{ m s}^{-1} \quad (\text{shown})$$

(d)



[B1] Before collision: u_A at 2.4 m s^{-1} and u_B at 0 m s^{-1}

[B1] shape: Smooth and continuous curvature, 0.48 m s^{-1} label

[B1] After collision: v_A at −1.44 m s^{-1} and u_B at 0 m s^{-1}